

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
ProgramName: B. Tech		Assignment Type: Lab	AcademicYear: 2025-2026
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Instructor(s)Name		1. Dr. Mohammed Ali Shaik 2. Dr. T Sampath Kumar 3. Mr. S Naresh Kumar 4. Dr. V. Rajesh 5. Dr. Brij Kishore 6. Dr Pramoda Patro 7. Dr. Venkataramana 8. Dr. Ravi Chander 9. Dr. Jagjeeth Singh	
CourseCode	24CS002PC215	CourseTitle	AI Assisted Coding
Year/Sem	II/I	Regulation	R24
Date and Day of Assignment		Time(s)	
Duration	2 Hours	Applicable to Batches	
AssignmentNumber: 3.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab 3: Prompt Engineering – Improving Prompts and Context Management Lab Objectives: - To understand how prompt structure and wording influence AI-generated code. - To explore how context (like comments and function names) helps AI generate relevant output. - To evaluate the quality and accuracy of code based on prompt clarity. - To develop effective prompting strategies for AI-assisted programming. Lab Outcomes (LOs): After completing this lab, students will be able to: - Generate Python code using Google Gemini in Google Colab. - Analyze the effectiveness of code explanations and suggestions by Gemini. - Set up and use Cursor AI for AI-powered coding assistance. - Evaluate and refactor code using Cursor AI features. - Compare AI tool behavior and code quality across different platforms.	03.08.2025 EOD	

Task Description#1

- Try 3 different prompts to generate a factorial function.

Expected Output#1

- Comparison of AI-generated code styles

Task Description#2

- Provide a clear example input-output prompt to generate a sorting function.

Expected Output#2

- Functional sorting code from AI

Task Description#3

- Start with the vague prompt “Generate python code to calculate power bill” and improve it step-by-step

Expected Output#3

- Enhanced AI output with clearer prompts

Task Description#4

- Write structured comments to help AI generate two linked functions (e.g., login_user() and register_user()).

Expected Output#4

- Consistent functions with shared logic

Task Description#5

- Analyzing Prompt Specificity: Improving Temperature Conversion Function with Clear Instructions

Expected Output#5

- Code quality difference analysis for various prompts

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

Evaluation Criteria:

Criteria	Max Marks
Factorial Function (Task#1)	0.5
Sorting Function (Task#2)	0.5
Vague Vs. Specific Prompting (Task #3)	0.5
Linked Functions (Task #4)	0.5
Temperature Conversion Function (Task #5)	0.5
Total	2.5 Marks

Task Description#1

- Try 3 different prompts to generate a factorial function.

Expected Output#1

- Comparison of AI-generated code styles

Prompt-1:

Generate a function that prints factorial of a given number

Code:

```
task1.py > main
1  def factorial(n):
2      if n < 0:
3          return "Factorial is not defined for negative numbers"
4      elif n == 0 or n == 1:
5          return 1
6      else:
7          result = 1
8          for i in range(2, n + 1):
9              result *= i
10         return result
11
12  def main():
13      Ctrl+L to chat, Ctrl+K to generate
14      try:
15          # Get input from user
16          user_input = input("Enter a number to calculate factorial: ")
17
18          # Convert to integer
19          number = int(user_input)
20
21          # Calculate factorial
22          result = factorial(number)
23
24          # Print the result
25          if isinstance(result, str):
26              print(result)
27          else:
28              print(f"Factorial of {number} is: {result}")
29
30      except ValueError:
31          print("Error: Please enter a valid integer")
32      except Exception as e:
33          print(f"An error occurred: {e}")
34
35  if __name__ == "__main__":
36      main()
37
```

Output:

```
Enter a number to calculate factorial: 5
Factorial of 5 is: 120
PS C:\Users\thoop\OneDrive\文档\AIAC_LAB_Lab3> & C:/Users/thoop/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thoop/OneDrive/文档/AIAC_LAB_Lab3/task1.py
Enter a number to calculate factorial: 8
Factorial of 8 is: 40320
```

Prompt 2:

Write a simple python function to find factorial of a number ,collect input from the console

Code:

```
task2.py > ...
1  def factorial(n):
2      if n < 0:
3          return "Error: Factorial is not defined for negative numbers"
4      elif n == 0 or n == 1:
5          return 1
6      else:
7          result = 1
8          for i in range(2, n + 1):
9              result *= i
10         return result
11
12 # Collect input from console
13 user_input = input("Enter a number to calculate factorial: ")
14
15 try:
16     number = int(user_input)
17     result = factorial(number)
18     print(f"Factorial of {number} is: {result}")
19 except ValueError:
20     print("Error: Please enter a valid integer")
21
```

Problems Output Debug Console Terminal Ports

```
6
PS C:\Users\thoop\OneDrive\文档\AIAC_LAB_Lab3> & C:/Users/thoop/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thoop/OneDrive/文档/AIAC_LAB_Lab3/task2.py
Enter a number to calculate factorial: 5
Factorial of 5 is: 120
PS C:\Users\thoop\OneDrive\文档\AIAC_LAB_Lab3> 
```

Prompt-3:

Create a simple python function that gives factorial of a number,use input from the programmer

```
task2.py > ...
1 def factorial(n):
2     """
3     Calculate the factorial of a given number.
4     """
5     if n < 0:
6         return "Error: Factorial is not defined for negative numbers"
7     elif n == 0 or n == 1:
8         return 1
9     else:
10        result = 1
11        for i in range(2, n + 1):
12            result *= i
13        return result
14
15 # Programmer provides the input number here
16 number = 5 # Change this number to calculate factorial of different numbers
17
18 # Calculate and display the factorial
19 result = factorial(number)
20 print(f"Factorial of {number} is: {result}")
21
```

Problems Output Debug Console **Terminal** Ports

/OneDrive/文档/AIAC_LAB/Lab3/task2.py
Factorial of 5 is: 120
PS C:\Users\thoop\OneDrive\文档\AIAC_LAB\Lab3> & C:/Users/thoop/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thoop/OneDrive/文档/AIAC_LAB/Lab3/task2.py
Factorial of 5 is: 120
PS C:\Users\thoop\OneDrive\文档\AIAC_LAB\Lab3>

Task Description#2

- Provide a clear example input-output prompt to generate a sorting function.

Expected Output#2

- Functional sorting code from AI

Prompt:

Generate a python code that takes a input list from the user and sort the list
Ascending order

Code along with output:

```
1 def sort_list_ascending():
2     """
3     Function to take input list from user and sort it in ascending order
4     """
5     try:
6         # Get input from user
7         print("Enter numbers separated by spaces (e.g., 5 2 8 1 9):")
8         user_input = input("Enter your list: ")
9
10        # Convert input string to list of integers
11        numbers = [int(x) for x in user_input.split()]
12
13        # Sort the list in ascending order
14        sorted_list = sorted(numbers)
15
16        # Display the results
17        print(f"Original list: {numbers}")
18        print(f"Sorted list (ascending): {sorted_list}")
19
20    except ValueError:
21        print("Error: Please enter valid numbers separated by spaces")
22    except Exception as e:
23        print(f"An error occurred: {e}")
24
25 def sort_list_with_builtin():
26     """
27     Alternative method using list.sort() method
28     """
29     try:
30         print("Enter numbers separated by spaces:")
31         user_input = input("Enter your list: ")
32
33         # Convert input string to list of integers
34         numbers = [int(x) for x in user_input.split()]
35
36         # Create a copy to avoid modifying original list
37         numbers_copy = numbers.copy()
38
39         # Sort the list in ascending order using sort() method
40         numbers_copy.sort()
41
42         print(f"Original list: {numbers}")
43         print(f"Sorted list (ascending): {numbers_copy}")
44
45    except ValueError:
46        print("Error: Please enter valid numbers separated by spaces")
47
48 # Main execution
49 if __name__ == "__main__":
50     print("=== List Sorting Program ===")
51     print("Method 1: Using sorted() function")
52     sort_list_ascending()
53
54     print("\n" + "-"*40 + "\n")
55
56     print("Method 2: Using sort() method")
57     sort_list_with_builtin()
58
59 Problems Output Debug Console Terminal Ports
Method 1: Using sorted() function
Enter numbers separated by spaces (e.g., 5 2 8 1 9):
Enter your list: 5 66 4 22 3 8 99 0 33
Original list: [5, 66, 4, 22, 3, 8, 99, 0, 33]
Sorted list (ascending): [0, 3, 4, 5, 8, 22, 33, 66, 99]
```

Task Description#3

- Start with the vague prompt “Generate python code to
- calculate power bill” and improve it step-by-step

prompt:

generate a python code to calculate powerbill containing previous units ,present units ,read input units from the console

code:

```
task3.py > ...
1 def calculate_power_bill():
2     """
3     Function to calculate power bill based on previous and present units
4     """
5     try:
6         # Get input from user
7         print("=== Power Bill Calculator ===")
8
9         # Get previous units
10        previous_units = float(input("Enter previous month units: "))
11
12        # Get present units
13        present_units = float(input("Enter present month units: "))
14
15        # Calculate units consumed
16        units_consumed = present_units - previous_units
17
18        # Calculate bill based on units consumed
19        if units_consumed <= 0:
20            print("Error: Present units should be greater than previous units")
21            return
22
23        # Define rate structure (example rates - can be modified)
24        if units_consumed <= 100:
25            rate_per_unit = 2.50
26        elif units_consumed <= 200:
27            rate_per_unit = 3.50
28        elif units_consumed <= 300:
29            rate_per_unit = 4.50
30        else:
31            rate_per_unit = 5.50
32
33        # Calculate total bill
34        total_bill = units_consumed * rate_per_unit
35
36        # Display the results
37        print("\n=== Bill Details ===")
38        print(f"Previous month units: {previous_units}")
39        print(f"Present month units: {present_units}")
40        print(f"Units consumed: {units_consumed}")
41        print(f"Rate per unit: ${rate_per_unit}")
42        print(f"Total bill: ${total_bill:.2f}")
43
44    except ValueError:
45        print("Error: Please enter valid numbers")
46    except Exception as e:
47        print(f"An error occurred: {e}")
48
49 def calculate_power_bill_with_tax():
50     """
51     Function to calculate power bill with tax included
52     """
53     try:
54         print("\n=== Power Bill Calculator with Tax ===")
55
56         # Get input from user
57         previous_units = float(input("Enter previous month units: "))
58         present_units = float(input("Enter present month units: "))
59
60         # Calculate units consumed
61         units_consumed = present_units - previous_units
62
63         if units_consumed <= 0:
64             print("Error: Present units should be greater than previous units")
65             return
66
67         # Calculate base bill
68         if units_consumed <= 100:
69             rate_per_unit = 2.50
70         elif units_consumed <= 200:
71             rate_per_unit = 3.50
72         elif units_consumed <= 300:
```

```
73     rate_per_unit = 4.50
74 else:
75     rate_per_unit = 5.50
76
77 base_bill = units_consumed * rate_per_unit
78
79 # Calculate tax (example: 10% tax)
80 tax_rate = 0.10
81 tax_amount = base_bill * tax_rate
82
83 # Calculate total bill with tax
84 total_bill = base_bill + tax_amount
85
86 # Display detailed results
87 print("\n=== Detailed Bill ===")
88 print(f"Previous month units: {previous_units}")
89 print(f"Present month units: {present_units}")
90 print(f"Units consumed: {units_consumed}")
91 print(f"Rate per unit: ${rate_per_unit}")
92 print(f"Base bill: ${base_bill:.2f}")
93 print(f"Tax {(tax_rate*100)%}: ${tax_amount:.2f}")
94 print(f"Total bill: ${total_bill:.2f}")
95
96 except ValueError:
97     print("Error: Please enter valid numbers")
98 except Exception as e:
99     print(f"An error occurred: {e}")
100
101 # Main execution
102 if __name__ == "__main__":
103     # Run basic power bill calculator
104     calculate_power_bill()
105
106     # Run power bill calculator with tax
107     calculate_power_bill_with_tax()
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```

Task Description#4

- Write structured comments to help AI generate two linked functions (e.g., login_user() and register_user()).

Prompt:

Generate simple two python linked functions one is for register user and login user for account creation

Code:


```

task4.py > ...
1  # Global dictionary to store user accounts
2  user_accounts = {}
3
4  def register_user():
5      """
6      Function to register a new user account
7      """
8      print("=== Account Registration ===")
9
10     # Get user input for account creation
11     username = input("Enter username for your account: ").strip()
12
13     # Check if username already exists
14     if username in user_accounts:
15         print("Error: Username already exists! Please choose a different username.")
16         return False
17
18     # Get password for account
19     password = input("Enter password for your account: ")
20
21     # Store user account credentials
22     user_accounts[username] = password
23
24     print("Account created successfully!")
25     print(f"Username: {username}")
26     print("You can now login with your account.")
27     return True
28
29  def login_user():
30      """
31      Function to login to existing user account
32      """
33      print("=== Account Login ===")
34
35     # Get login credentials
36     username = input("Enter your username: ").strip()
37     password = input("Enter your password: ")
38
39     # Check if account exists and password matches
40     if username in user_accounts and user_accounts[username] == password:
41         print("Login successful!")
42         print(f>Welcome to your account, {username}!")
43         return True
44     else:
45         print("Invalid username or password.")
46         print("Please check your account credentials.")
47         return False
48
49  # Main execution for account management
50  if __name__ == "__main__":
51      while True:
52          print("\n=== Account Management System ===")
53          print("1. Create Account")
54          print("2. Login to Account")
55          print("3. Exit")
56
57          choice = input("Enter your choice (1-3): ").strip()
58
59          if choice == '1':
60              register_user()
61          elif choice == '2':
62              login_user()
63          elif choice == '3':
64              print("Thank you for using Account Management System!")
65              break
66          else:
67              print("Invalid choice! Please try again.")

```

Output:

```
=== User Registration ===
Enter username: 2403a51057
Enter password: hary66
Confirm password: hary66
Enter email: 57@gmail.com
Enter full name: HarichandanaThoopukari
Enter phone number: 1236661230

=== Registration Successful ===
Username: 2403a51057
Email: 57@gmail.com
Full Name: HarichandanaThoopukari
Phone: 1236661230
Registration Date: 2024-01-01
Status: Active

=== User Management System ===
1. Register User
2. Login User
3. Display All Users
4. Exit

Enter your choice (1-4): 2
=== User Login ===
Enter username: 2403a51057
Enter password: hary66

=== Login Successful ===
Welcome back, HarichandanaThoopukari!
Username: 2403a51057
Email: 57@gmail.com
Full Name: HarichandanaThoopukari
Phone: 1236661230
Registration Date: 2024-01-01
Status: Logged In
Last Login: Current Session
```

Task Description#5

- Analyzing Prompt Specificity: Improving Temperature Conversion Function with Clear Instructions

Prompt1:

Write a Python function to convert temperature from Celsius to Fahrenheit or vice-versa as for the user`s selection

Code:

```

task5.py > ...
1  def convert_temperature():
2      """
3      Converts temperature between Celsius and Fahrenheit based on user input.
4      """
5      try:
6          print("Temperature Converter")
7          temp = float(input("Enter the temperature value: "))
8          unit = input("Is this in Celsius or Fahrenheit? (C/F): ").strip().upper()
9
10         if unit == 'C':
11             converted = (temp * 9/5) + 32
12             print(f"{temp}°C is {converted:.2f}°F")
13         elif unit == 'F':
14             converted = (temp - 32) * 5/9
15             print(f"{temp}°F is {converted:.2f}°C")
16         else:
17             print("Invalid unit. Please enter 'C' for Celsius or 'F' for Fahrenheit.")
18     except ValueError:
19         print("Invalid input. Please enter a numeric temperature value.")
20
21     # Call the function if this script is run directly
22     if __name__ == "__main__":
23         convert_temperature()
24

```

Problems Output Debug Console Terminal Ports

```

p' '--' 'c:\Users\thoop\OneDrive\文档\AIAC_LAB_\Lab3\task5.py'
Temperature Converter
Enter the temperature value: 15
Is this in Celsius or Fahrenheit? (C/F): c
15.0°C is 59.00°F
PS C:\Users\thoop\OneDrive\文档\AIAC_LAB_\Lab3>

```

Prompt2:

Write a Python function to convert temperature from Celsius to Fahrenheit or vice-versa as for the user's selection.

Code along with output:

```

task5_1.py > ...
1  def convert_temperature():
2      """
3      Converts temperature between Celsius and Fahrenheit based on user's selection.
4      """
5      try:
6          print("Temperature Converter")
7          print("Select conversion type:")
8          print("1. Celsius to Fahrenheit")
9          print("2. Fahrenheit to Celsius")
10         choice = input("Enter 1 or 2: ").strip()
11
12         if choice == '1':
13             celsius = float(input("Enter temperature in Celsius: "))
14             fahrenheit = (celsius * 9/5) + 32
15             print(f"{celsius}°C is {fahrenheit:.2f}°F")
16         elif choice == '2':
17             fahrenheit = float(input("Enter temperature in Fahrenheit: "))
18             celsius = (fahrenheit - 32) * 5/9
19             print(f"{fahrenheit}°F is {celsius:.2f}°C")
20         else:
21             print("Invalid selection. Please enter 1 or 2.")
22     except ValueError:
23         print("Invalid input. Please enter a numeric temperature value.")
24
25 if __name__ == "__main__":
26     convert_temperature()

```

Problems Output Debug Console **Terminal** Ports

```

Select conversion type:
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
Enter 1 or 2: 1
Enter temperature in Celsius: 35
35.0°C is 95.00°F
PS C:\Users\thoop\OneDrive\文档\AIAC_LAB_\Lab3> 

```

+ v ...

power
Python
Python